CYBERSENTINEL COPILOT

Detailed architecture and numbered end-to-end flow

Each stage shows ownership, processing, dependencies, output, and the next transition

STEP / OWNER	PROCESSING AND DECISIONS	DATA / MODEL / EXTERNAL DEPENDENCIES	OUTPUT AND NEXT STATE
0 · Offline foundation Preparation scripts	**Processing** Audit and normalize CIC-IDS2017 Map labels to nine attack families Seed 291 incident summaries and FTS5 Train classifier and benign-only anomaly model	**Dependencies** 8 Parquet files · 78 columns SQLite · scikit-learn · joblib Curated MITRE/NIST documents	**Output** 2,313,810 SQLite flows 65-feature classifier bundle IsolationForest bundle Synapt knowledge corpus
1 · Analyst + API entry Dashboard / FastAPI	**Processing** Collect record ID, query, family, protocol, or raw flow Validate types, bounds, length, and non-empty request Dispatch POST /investigate/flow	**Dependencies** index.html FlowInvestigationRequest FastAPI route wiring	**Output** Validated request HTTP 422 on contract failure LangGraph initial state
2 · Guardrail guardrail_agent	**Processing** Detect prompt injection and destructive instructions Block unauthorized automated containment Check cybersecurity relevance for query-only requests	**Dependencies** Guardrail regex policy Domain relevance patterns No external side effects	**Output** guardrail_status: passed/rejected Violation reason when blocked Sanitized path to triage
3 · Flow triage + RAG flow_triage_agent	**Processing** Load exact record or validate raw telemetry Enforce real 404 for missing explicit record Run BM25 and semantic channels; RRF, filter, self-exclude	**Dependencies** SQLite flows · FTS5 summaries HistoricalIncidentRAG Synapt semantic retrieval	**Output** target_flow or unassessed state similar_flows target_not_found flag
4 · ML analysis ml_analysis_agent	**Processing** Extract 65 traffic features Predict family, confidence, risk = 1 − P(benign), severity Calculate top five deviations and anomaly score	**Dependencies** HistGradientBoostingClassifier Benign feature baseline IsolationForest + threshold	**Output** ml_prediction anomaly_prediction MITRE tactic mapping
5 · Knowledge retrieval synapt_knowledge_agent	**Processing** Keep rejected text outside provider boundary Sanitize analyst context and ask a family-specific question Retrieve procedures, ATT&CK indicators, verification steps	**Dependencies** SynaptClient · TLS/auth/JSON-RPC MITRE/NIST corpus Deterministic local fallback	**Output** synapt_evidence synapt_analysis Graceful disconnected status
6 · Risk prioritization prioritization_agent	**Processing** Combine family, risk, anomaly, and known context Apply explicit P1–P4 attack-family policy Record unknown asset and host identity context	**Dependencies** determine_incident_priority Classification + anomaly outputs Optional asset criticality	**Output** priority_level and code Policy rationale missing_context list
7 · Response synthesis response_recommendation_agent	**Processing** Assemble investigation sequence and response drafts Attach record/model/provider provenance Disclose dataset, metadata, and automation limits	**Dependencies** Prediction · priority · similar incidents Synapt analysis · MITRE mappings NIST-aligned static response template	**Output** investigation_steps recommended_response source_provenance + limitations
8 · Output validation judge_review_agent	**Processing** Reject prohibited automatic-action language first Score grounding, action safety, NIST alignment, relevance Parse model JSON when provider-backed judge is available	**Dependencies** ResponseJudge SynaptDeepEvalLLM adapter Fixed rubric fallback	**Output** pass_verdict composite + dimension scores critique + unsupported claims
9 · Present + improve Dashboard / analyst	**Processing** Render the complete evidence package Human verifies findings and approves any response Submit rating, verdict, notes, optional corrected family	**Dependencies** FlowInvestigationResponse analyst_feedback table Human review gate	**Output** Analyst decision Pending feedback record Accepted/rejected improvement input

SUPPORTING PATHS THAT SHARE THE SAME DATA, MODELS, AND PROVIDER

Incident search
Structured SQL + FTS5 BM25
+ Synapt semantic channel + RRF

Conversational copilot
Guardrails + optional flow context
+ classifier + MITRE + Synapt/local reply

Feedback and improvement
Submit → human review → holdout
+ candidate artifact + benchmark report

Evaluation evidence
40 fixtures + provider adapter
+ local safety gate + JSON reports

SOURCE-REVIEWED BOUNDARIES

Metadata
Local export lacks IP, port, Flow ID, timestamp fields.

Retrieval
Sparse fused sets are padded with random corpus records.

Evaluation / judge
Benchmark score and offline judge fallback use fixed defaults.

Improvement / health
Candidate copies active model; Synapt health is config-based.

Solid arrows: runtime transition · dashed arrows: dependency input · all response actions require human approval